

Opening up the subsoil: Subsoiling and bio-subsoilers to remediate subsoil compaction in three fodder crop rotations on a sandy loam soil

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ABSTRACT

The intensification of modern arable farming has brought on a steady increase of the risk of soil compaction, while simultaneously pushing the impact deeper into the soil profile. In the subsoil, outside the reach of regular tillage operation, it can become exceedingly difficult to deal with compacted layers. Commonly, farmers turn towards mechanical remediation strategies, like subsoiling, to deal with this issue. However, the numerous drawbacks that come with these operations have prompted the search for more sustainable alternatives, like bio-subsoilers, which are deep-rooting plant species with the capacity to penetrate compacted subsoil layers. This study aimed at investigating both mechanical and biological remediation strategies in a side-by-side comparison. The effectiveness of subsoiling and the use of two deep rooting (cover) crops (i.e. alfalfa and fodder radish) as remediation measures to alleviate subsoil compaction was assessed in terms of soil quality indicators, like penetration resistance, bulk density, porosity, plant available water capacity, air capacity and air permeability at 35 and 45 cm depth, and the above- and belowground growth of the main crop. In addition, a combination of both subsoiling and use of deep rooting (cover) crops was included to evaluate the potential role of the latter in protecting the mechanically loosened subsoil from recompaction. This complex comparison was done in three distinct maize-based fodder crop rotations (rotation 1: maize monoculture; rotation 2: maize/winter cereal rotation with fodder radish as cover crop after winter cereal; rotation 3: maize/winter cereal/alfalfa rotation). The study field (sandy loam soil) was selected based on the presence of a highly compacted layer at the interface between topsoil and subsoil (30–50 cm). The experiment demonstrated that subsoiling is an effective remediation measure to disrupt heavily compacted layers in the upper-subsoil, at least in the short-term. The crop response was however highly variable and only positive when the opportunity of reaching the subsoil proved to be advantageous for the crop (i.e. dry season). Besides its high degree of variability between years, the effects also proved to be very short-lived under a standard arable fodder crop rotation. Including deep rooting (cover) crops, like fodder radish and alfalfa, and limiting the initial subsoil disruption helped to partially stabilize the disrupted subsoil and reduce recompaction. Fodder radish and alfalfa penetrated the compacted subsoil without previous disruption by subsoiling. However, this did not lead to a significant improvement of the measured soil physical parameters.

1. Introduction

The subsoil can function as an extensive reservoir of water and nutrients (Colombi et al., 2018; Kautz et al., 2013; Lynch and Wojciechowski, 2015; Wiesmeier et al., 2013). For example, it can take up excess water during intense rainfall and thereby reduce the risk of flooding and erosion (Alaoui et al., 2018), and hold it longer during prolonged dry spells (Gaiser et al., 2012; Kirkegaard et al., 2007;

McKenzie et al., 2009). However, the past few decades have seen an increased strain on the subsoils' capacity to fulfil these ecological and agricultural services (Keller et al., 2019). The use of ever heavier farming machinery and the increased frequency of field traffic under wet, and thus weaker soil conditions (Gut et al., 2015; Koolen and Kuipers, 1983; Saffih-Hdadi et al., 2009; STATBEL, 2022), have seriously increased the risk of compaction of deeper soil layers (Keller and Or, 2022; Lamandé and Schjøning, 2011a, 2011b). These more

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compacted subsoils constrain water infiltration and storage (Keller et al., 2019; Soane et al., 1980) and increase the mechanical impedance to deeper root growth (Bengough et al., 2006; Lynch and Wojciechowski, 2015; Materechera et al., 1991).

While a compacted topsoil can more easily be remediated by natural processes (e.g. swelling-shrinking, freezing-thawing) (Dexter, 1991) and by regular tillage operations (Arvidsson and Håkansson, 1996), compacted subsoils will be much harder to handle (Keller et al., 2021; Munkholm et al., 2005; Schneider et al., 2017) and soil compaction can therefore become cumulative and semi-permanent (Arvidsson and Keller, 2007; Défossez et al., 2003; Håkansson, 1985; Håkansson and Reeder, 1994; Sharratt et al., 1998). Deep tillage operations, like subsoiling (i.e. loosening of the soil by lifting up the profile without inversion or mixing, also referred to as deep ripping or deep chiseling), have been shown to effectively loosen up compacted subsoil layers in a wide variety of soil types (Elsen et al., 2014; Schneider et al., 2017; Soane et al., 1987). The level of soil disturbance depends on soil conditions during subsoiling and subsoiler characteristics, like the attachment of wings, spacing between the tines and tine inclination (Hamza and Anderson, 2003; Raper, 2005; Spoor, 2006; Spoor and Godwin, 1978). However, the impact on crop production will not only depend on the level of soil disturbance. Other factors, like subsequent weather conditions, play a crucial role in the crop response. Dry spells, for example, increase the potential benefits of subsoiling (Olesen and Munkholm, 2007; Schneider et al., 2017). Subsoiling also has other important drawbacks, like the high fuel consumption (Hamza and Anderson, 2008; Kichler et al., 2011) and high risk of recompaction (Canarache et al., 2000; Munkholm et al., 2005; Spoor et al., 2003). The latter makes it paramount that subsoiling be followed by soil conserving measures, like on-land ploughing (Munkholm et al., 2005; Olesen and Munkholm, 2007), correctly lowering tyre inflation pressure (Soane et al., 1986), limiting field traffic (Elsen et al., 2014; Munkholm et al., 2005; Soane et al., 1986) and changing tillage from plough to disk harrow (Tessier et al., 1997). Management strategies that help to stabilize the disturbed subsoil structure could also be used against recompaction. Soil organic matter (e.g. Getahun et al., 2018; Gill et al., 2009; Piccoli et al., 2022) and gypsum (mostly for clay-rich soils) (e.g. Hamza and Anderson, 2005; Löfkvist, 2005) could be incorporated during subsoiling to work as binding or cementing agents. Roots of subsequent crops have a similar potential, both directly by entanglement of the soil particles, as indirectly by bringing in organic matter and the associated microbial activity (Angers and Caron, 1998; De Baets et al., 2008; Morel et al., 1991; Tisdall and Oades, 1982; Tisdall and Oades, 1979). However, to the best of our knowledge, research on the stabilizing impact of deep rooting plant species after subsoiling has been rather limited. Löfkvist (2005) did observe a prolonged positive impact of subsoiling on hydraulic conductivity at 35 cm depth by alfalfa (*Medicago sativa*) after three years.

Some deep rooting plant species additionally have the capacity to penetrate compacted subsoil layers and thereby help mitigate the negative consequences associated with subsoil compaction (Cresswell and Kirkegaard, 1995; Elkins, 1985). This process has been referred to as biological drilling (Cresswell and Kirkegaard, 1995) or biological subsoiling (Löfkvist, 2005; Pulido-Moncada et al., 2020) and plant species that have this potential could be referred to as bio-subsoilers (Pulido-Moncada et al., 2020). The penetration of compacted soil layer creates stable biopores (Hagedorn and Bundt, 2002; Schäffer et al., 2007) that can be recycled by the roots of subsequent crops, to reach the otherwise less accessible subsoil resources (Cresswell and Kirkegaard, 1995; Kautz, 2015; Landl et al., 2019; Perkons et al., 2014; Uteau et al., 2013). Previous studies have observed marked differences between plant species in their capacity to act as bio-subsoilers. In general, tap-rooted dicotyledons will have a higher potential than fibrous-rooted monocotyledons (Chen and Weil, 2010; Clark and Barraclough, 1999; Colombi et al., 2017b; Han et al., 2015; Löfkvist, 2005; Materechera et al., 1991). This difference can probably be explained by their on

average larger root diameter (Materechera et al., 1992) and more vertically directed root growth (Beck-Broichsitter et al., 2020). For example, tap-rooted crops like the fodder crop, alfalfa, and the cover crop, fodder radish (*Raphanus sativus*), have a clear capacity to grow through compacted subsoil layers (Chen and Weil, 2010; Han et al., 2015; Kautz et al., 2010; Pulido-Moncada et al., 2020; Uteau et al., 2013), to improve the degraded soil structure (Abdollahi and Munkholm, 2014; Beck-Broichsitter et al., 2020; Chen et al., 2014; Hudek et al., 2022; Meek et al., 1990; Mitchell et al., 1995; Pulido-Moncada et al., 2021a) and to stimulate deeper root exploration of subsequent crops (Chen and Weil, 2011; Gaiser et al., 2012; Guaman et al., 2016; Stirzaker et al., 1996; Williams and Weil, 2004). It should be noted though that the impact of fodder radish is much more variable than that of alfalfa, which can be partially explained by differences between the utilized varieties (Abdollahi and Munkholm, 2014; Chen and Weil, 2010; Guaman et al., 2016; Piccoli et al., 2022; Pulido-Moncada et al., 2021b, 2020; Welch et al., 2016).

Unlike studies that looked at biological (e.g. bio-subsoilers) and mechanical (e.g. subsoiling) remediation strategies separately, experiments that looked at both strategies in a side-by-side comparison (Blanco-Canqui and Wortmann, 2020) are rare. Löfkvist (2005) evaluated the remediation effect of alfalfa and of subsoiling, and the influence of the combination of both on recompaction of a silty clay soil. Inagaki et al. (2021) looked at fodder radish and chiseling on a sandy clay soil, but only investigated the topsoil (0–30 cm). Guamán Sarango (2015) similarly looked at fodder radish and subsoiling of a sandy loam soil, but did not assess the impact on recompaction.

This field experiment aimed to give a side-by-side comparison of subsoiling and two deep rooting (cover) crops (i.e. alfalfa and fodder radish), based on both short- and longer-term (i.e. one year and > one year, respectively) effectiveness as remediation measures to deal with subsoil compaction (i.e. soil physical quality) and the above- and belowground crop response. For subsoiling we hypothesized a short-term loosening effect with a general improvement of the soil physical quality, increased subsoil root exploration and increase in crop yield, but with quick recompaction. We expected that the deep rooting crops would have a far more sustainable impact. The perennial crop, alfalfa, was hypothesized to have a larger impact than the cover crop, fodder radish. In addition, a combination of both subsoiling and deep rooting (cover) crops was included to evaluate the potential role of the latter in protecting the mechanically loosened subsoil from recompaction. We hypothesized that crop choice and crop rotation play an important role in reducing recompaction after subsoiling and that the loosened subsoil can be stabilized by sowing deep rooting (cover) crops immediately after subsoiling.

2. Material and methods

2.1. Study site

The experimental site (50°59'10.2"N 3°46'57.9"E) was located in the municipality of Melle, Belgium. The study area has a temperate maritime climate with mild winters and warm summers (Kottek et al., 2006). The mean annual temperature is 10.9 °C and mean annual precipitation is 849.1 mm (1991–2020, Royal Meteorological Institute of Belgium, RMI). The Belgian soil map converted to a World Reference Base legend designates the field as Haplic Cambisol (Dondeyne et al., 2014; IUSS Working Group WRB, 2007). Table 1 presents the particle size distribution, organic carbon content, pH and observed soil horizons of the topsoil (0–30 cm) and upper subsoil (30–60 cm). These analyses were performed on six different composite samples, two per block (three subsamples, collected in the control sub-plots of each experimental treatment at positions 1 and 3 – see Fig. 1). The particle size distribution was determined with the sieving, sedimentation and pipette method (AFNOR - NF X31–107; Gee and Bauder, 1986). Both sampled soil layers can be designated as sandy loam according to the USDA classification.

Table 1
Average particle size distribution, organic carbon content and pH ± standard deviation (n = 6), the resulting texture class (USDA) and the observed soil horizons for the depth intervals 0–30 and 30–60 cm.

Depth (cm)	Sand (g kg ⁻¹)	Silt (g kg ⁻¹)	Clay (g kg ⁻¹)	Texture class (USDA)	Organic carbon content ^a (g kg ⁻¹)	pH (-)	Horizon
0–30	566.8 ± 41.2	371.7 ± 41.9	61.3 ± 6.7	Sandy loam	9.0 ± 0.6	6.6 ± 0.2	Ap ^b
30–60	548.3 ± 41.6	390.3 ± 37.4	61.2 ± 5.1	Sandy loam	< 4.9	6.9 ± 0.2	C ^c , Cg ^d

^a Detection limit organic carbon content: 4.9 g kg⁻¹
^b Ap: an A horizon (i.e. mineral horizon (≤17% organic carbon content) near the surface of the soil) that has been disturbed by human activity, such as mixing of the upper-soil by ploughing.
^c C: a C horizon is a mineral horizon that is comparatively unaffected by the pedogenic processes.
^d Cg: a C horizon characterized by dull, greyish colours and/or prominent mottling within 50 cm of the soil surface, indicating permanent or periodic intense reduction.

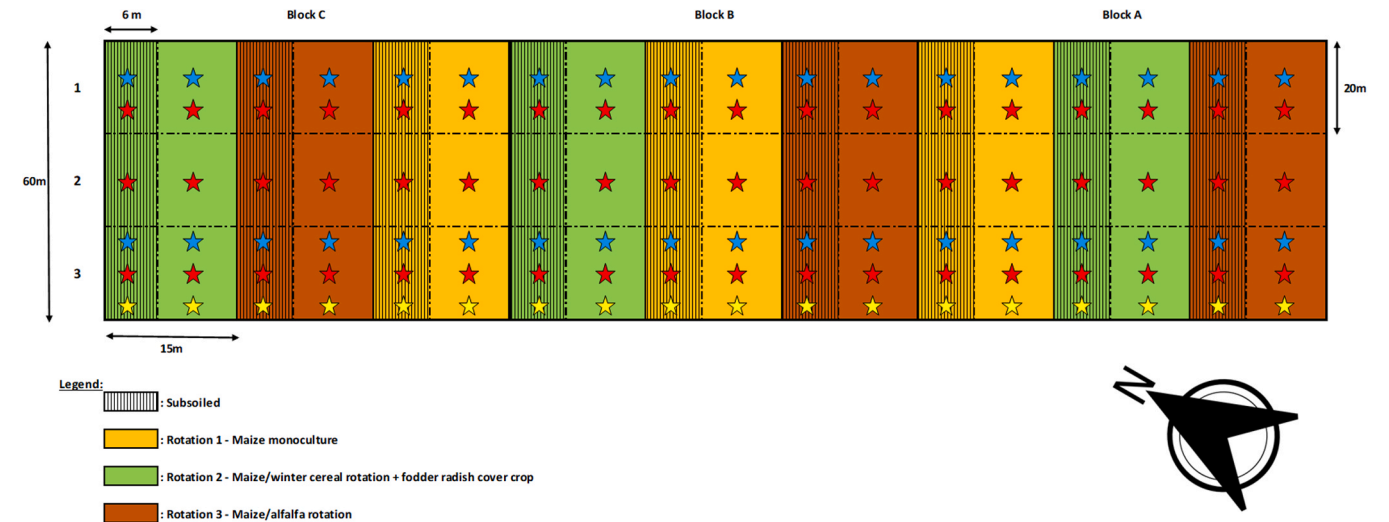


Fig. 1. Experimental design with three blocks (A, B and C). The stars represent the approximate locations of the experimental measurements: penetration resistance and the sampling of undisturbed soil cores on which bulk density, air capacity, plant available water capacity and air permeability were determined (blue), experimental harvesting (red) and the root number density and distribution (yellow).

The soil organic carbon content was determined after dry combustion (ISO 10694:2008) and the pH by water extraction (ISO 10390:2021). The soil horizons were classified in a series of profile pits (80 cm depth) throughout the experimental field.

Most soil properties presented in Table 1 were relatively homogeneous in across the experimental blocks. The biggest differences were observed for organic carbon content and the subsoil horizons. The organic carbon content was 8.6 g kg⁻¹ in block B, 8.7 g kg⁻¹ in block C and 9.7 g kg⁻¹ in block A, all below the target range for sandy loam arable fields in Belgium (10.0–11.6 g kg⁻¹; BDB and UGent, 2010). In blocks A and B we observed an undifferentiated C horizon (i.e. mineral horizon with ≤17% organic carbon content that is comparatively unaffected by the pedogenic processes) in the upper-subsoil (30–60 cm). Within block C the upward boundary of the Cg horizon (i.e. characterized by dull, greyish colours and/or prominent mottling within 50 cm of the soil surface, indicating permanent or periodic intense reduction) varied considerably. In the most northern sections of block C (see Fig. 1), the mottling could be observed up until the boundary of the topsoil (30 cm).

2.2. Experiment

The study field was selected based on the presence of a highly compacted layer at the interface between topsoil and subsoil (30–50 cm). This was determined with a hand-held penetrometer (Royal Eijkelkamp, the Netherlands) which showed penetration resistance values (measured at field capacity (i.e. 10 kPa matric suction) on 19 February, 2019) far above the standard threshold values for root

growth of 2–3 MPa (Bengough et al., 2011) (Fig. 2). Repeated penetration resistance measurements discerned a relatively even spread of the

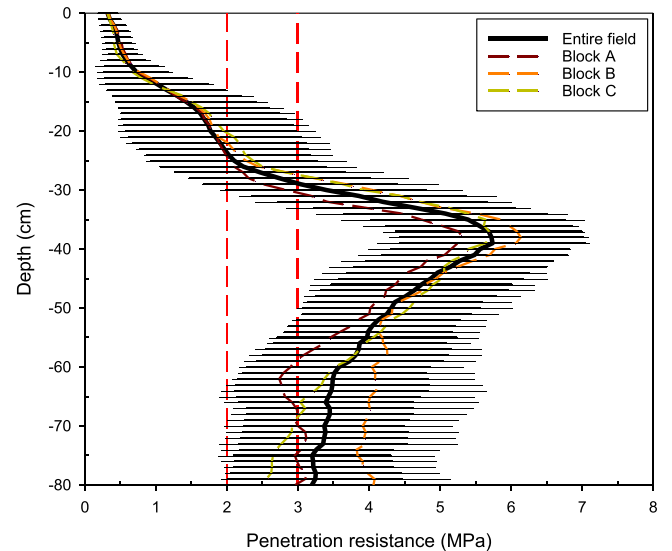


Fig. 2. Average penetration resistance (0–80 cm) ± standard deviation (horizontal black lines – n = 210) and the average penetration resistance for all three experimental blocks (A: brown, B: orange, C: green) before the establishment of the experiment. The dashed vertical straight lines indicate the threshold values for the inhibition of root growth (2 and 3 MPa).

compacted subsoil layer across the field, with the heaviest subsoil compaction not being confined to the much more frequently trafficked headlands (Augustin et al., 2020).

On February 22nd, 2019, a scan of the experimental field was made with a DUALEM-21S sensor which employs electromagnetic induction (EMI) to map the in-field variation of apparent electrical conductivity (ECa) of the soil profile. The scanning had an in-line resolution of ~ 0.2 m, with a 2 m interval between the lines of scanning. Details on the DUALEM-21S scanner can be found in De Smedt (2013). An ECa map (Fig. 3) made by interpolation with the Surfer® software (Golden Software, LLC) showed marked differences between different sections of the experimental field. The northern section was characterized by high ECa values, which gradually decreased towards the south-east. These marked differences can most likely be related to differences in moisture content, given the homogeneity of the field (Section 2.1). The gravimetric moisture content (determined on composite samples on the day of the EMI scan) and the associated matric suction (derived from the water retention curve measured at the start of the experiment – in parentheses hereafter) of the topsoil (0–30 cm) varied from 0.19 g g^{-1} (17.35 kPa) in the south over 0.21 g g^{-1} (13.82 kPa) in the center to 0.25 g g^{-1} (8.82 kPa) in the north. While on the Belgian soil map (Fig. 3) the south and center of the field are defined as relatively dry (well drained), the north is classified as relatively wet, especially in winter when the groundwater table can be shallow (imperfect drainage) (Van Ranst and Sys, 2000). This corroborates with rust mottles we observed below 80 cm depth in the southern part of the field (block A), at shallower depth (< 80 cm) in the center from block B onwards and already at 40 cm depth by the most northern subplots of block C.

The experiment was set up as a split-plot design with crop rotation as whole-plot factor and subsoiling as sub-plot factor in 3 replications (Fig. 1). Crop rotation consisted of three fodder crop rotations, while subsoiling entailed subsoiled and not-subsoiled, further also referred to as control. This combination was repeated in three blocks, A, B and C, with C being located most to the north and A most to the south. The three experimental crop rotations were: a maize monoculture with winter rye as a cover crop (rotation 1), a maize/winter cereal rotation with fodder radish as cover crop after winter cereal (rotation 2) and a

maize/winter cereal/alfalfa rotation (rotation 3).

Rotation 1 had four maize crops during this period, with winter rye as a cover crop in the winter of 2018–2019, 2021–2022 and 2022–2023. In rotation 2 the maize cultivated in 2019 and 2021 was alternated with winter wheat in 2020 and winter barley in 2022. Both winter cereals were followed by the deep rooting cover crop, fodder radish. In rotation 3 the maize crop in 2019 was followed by winter wheat in 2020 to allow for the necessary earlier sowing date of alfalfa. The perennial crop was maintained for two years until the start of 2023. The deep rooting (cover) crops, alfalfa and fodder radish, were sown in 2020, right after subsoiling. The crop sequence and the most important field operations in all three rotations over the experimental period (2019–2022) are further summarized in Supplement 1.

All crops were fertilized according to good agricultural management practices. Spring crops (i.e. maize) were fertilized with a combination of cattle slurry and mineral fertilizers. About one week before the sowing of the crops in spring, cattle slurry was applied at a rate of $170 \text{ kg N ha}^{-1} \text{ y}^{-1}$ (i.e. the maximum allowed dose according to the current Manure Decree). The surface application by means of a slurry injector was immediately followed by incorporation with a (disc) cultivator (0–10 cm). A few days later the soil was tilled to a depth of 30 cm using a conventional plough. Mineral N fertilizers were applied on top of the slurry to meet the crop requirements. Finally, the seedbed was prepared using a rotary harrow with a cultivation depth of around 15 cm. Both the winter crops (i.e. winter wheat and barley) and the perennial crop alfalfa were sown in fall. The sowing was preceded by conventional ploughing (30 cm depth) after which a rotary harrow-sowing combination was used. Solely mineral N fertilizers were applied to the winter crops in two (barley) or three (wheat) splits during the growing season while mineral K fertilizer was applied to alfalfa in several splits after the different cuts at a total rate of $300 \text{ kg K}_2\text{O ha}^{-1} \text{ year}^{-1}$.

As indicated in Fig. 1, each rotation plot was partly subsoiled; this was done with the PROFUNDUS Z® subsoiler constructed by Steeno Ltd. (Belgium) (Supplement 2). The subsoiler was used in the standard cultivation direction for two passages at the left side of each plot, resulting in a subsoiled sub-plot with a width of 6 m and a non-subsoiled, or control, sub-plot of 9 m width. The subsoiling depth was

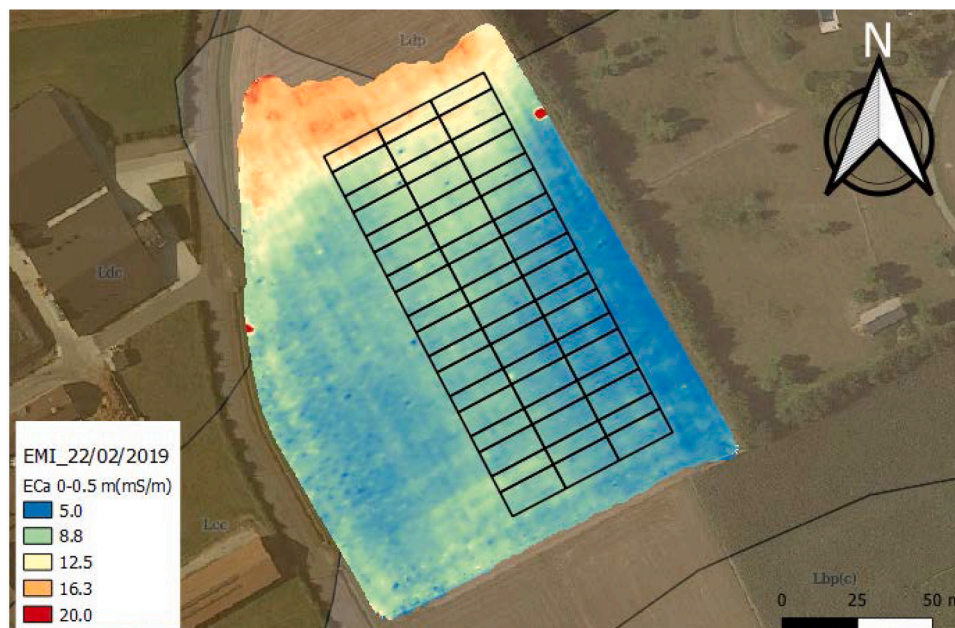


Fig. 3. Experimental plan overlaid onto the interpolated results of the electromagnetic induction (EMI) soil scan (DUALEM-21S sensor – PRP1), conducted on February 2nd 2019, which gives the apparent electrical conductivity (ECa – mS m^{-1}) of the top 0.5 m of the soil profile and the Belgian soil map, which divides the experimental field between the soil types, Ldp (i.e. a moderately wet sandy loam soil without profile development) in the northern edge and Lcp (i.e. a moderately dry sandy loam soil without profile development) in the rest.

consistently 50 cm. This depth was chosen so as to disrupt the compacted soil layer around 30–50 cm. Rotation 1 was subsoiled with seven tines spaced 45 cm apart on October 29th 2019 at rather dry soil conditions (Fig. 4). Under these conditions, the tractor (New Holland T7270) did not have too much difficulty to loosen up the soil down to 50 cm depth. When subsoiling in rotations 2 and 3 on August 31st 2020, however, soil conditions were significantly drier. The resulting increase in soil strength made it impossible to subsoil down to 50 cm depth with the previously used subsoiler-tractor combination. Therefore, the three front row tines on the subsoiler were removed and subsoiling was continued with the four back row tines remaining (Supplement 2-b), now with 90 cm spacing. For this reason, a comparison focusing only on the effects of the different crops of the tested rotations became unattainable. Further discussion of the experimental results invariably had to account the differing levels of subsoil disruption (rotation 1 vs. rotations 2 and 3).

Fig. 4 shows the soil moisture contents at subsoiling and the optimal soil moisture content for tillage, as defined by Dexter and Bird (2001), this is “the water content at which tillage produces the greatest proportion of small aggregates”. These authors located this condition at the inflection point ($w_{\text{inflection}}$ – i.e. water content at which air is entering a draining soil most rapidly) of the water retention curve, that could be determined with following equation:

$$w_{\text{inflection}} = (w_s - w_r) \left[1 + \frac{1}{m} \right]^{-m} + w_r$$

with w_s gravimetric water content at saturation, w_r residual gravimetric water content and m as shape parameter of the van Genuchten equation (Mualem, 1976). The conditions during subsoiling for rotation 1 (gravimetric water content (associated matric suction) – 0–10 cm: 0.16 g g⁻¹ (2.94 kPa); 10–30 cm: 0.17 g g⁻¹ (2.30 kPa); 30–60 cm: 0.15 g g⁻¹ (3.04 kPa)) were much closer to the optimal water content (gravimetric water content (associated matric suction) – 0–10 cm: 0.21 g g⁻¹ (1.37 kPa); 10–30 cm: 0.21 g g⁻¹ (1.42 kPa); 30–60 cm: 0.14 g g⁻¹ (3.33 kPa)) than the conditions for rotation 2 and 3 (gravimetric water content (associated matric suction) – 0–10 cm: 0.11 g g⁻¹ (5.59 kPa); 10–30 cm: 0.07 g g⁻¹ (22.06 kPa); 30–60 cm: 0.10 g g⁻¹ (6.86 kPa)). The associated matric potential was derived from the water retention curve measured at the start of the experiment.

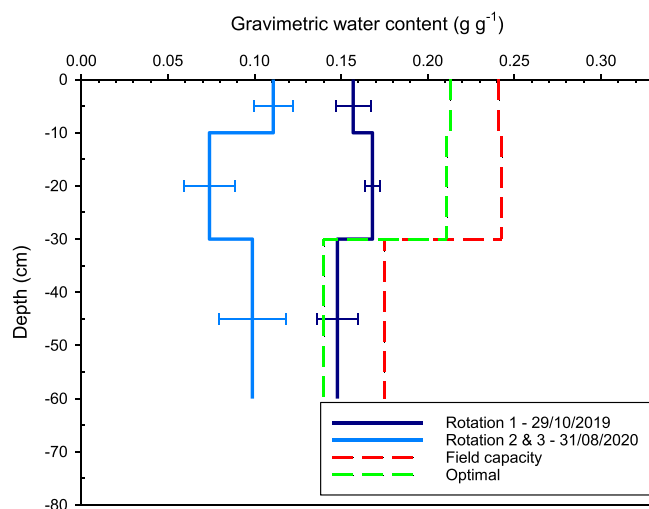


Fig. 4. Average gravimetric water content during the two experimental subsoiling operations at 29/10/2019 (Rotation 1) and 31/08/2020 (Rotation 2 and 3) $\pm$ standard deviation ($n = 6$) for three depth intervals (0–10, 10–30 and 30–60 cm). The given field capacity (–10 kPa - red) and the moisture content optimal for tillage (green), as defined by Dexter and Bird (2001), was determined for the pre-subsoiled conditions.

2.3. Weather conditions

A large portion of the experimental period was characterized by unseasonal weather conditions (Supplement 3). Periods of prolonged drought and high maximum temperatures were followed by months with extreme rainfall. Late spring and early summer of 2020 were relatively dry and had high peak temperatures. 2021, on the other hand, was relatively wet, especially in April and May, and had lower maximum temperatures. The summer of 2022 could be characterized as extremely dry, with only a few days of rain in July and August, and hot. September 2022 did experience the highest rainfall of the entire experimental period. The rainfall in the month immediately preceding the subsoiling of rotation 1 (29/10/2019) was normal for the time of year, whereas the subsoiling of rotation 2 and 3 (31/08/2020) was done after a prolonged period of relatively low rainfall.

2.4. Soil physical parameters

Penetration resistance was measured with a hand-held penetrometer (Royal Eijkkelkamp, the Netherlands) down to 80 cm depth. The cone had a 1 cm² base area, a 11.28 mm nominal diameter and a 60° top angle. During the experimental period the penetration resistance was measured on five different occasions for rotations 1 and 2 and three for rotation 3. The timing (both exact and relative to the subsoiling operation and sowing of the respective deep rooting (cover) crop) of all the measurements per experimental treatment is presented in Supplement 4. The measurements per sub-plot were consistently repeated on one transect (between 75 and 150 cm long depending on measurement date) perpendicular to the cultivation and subsoiling direction. The spacing between two measurements was 7.5 or 15 cm. This allowed us to draw contour plots of penetration resistance over the soil profile.

All laboratory measurements were performed on undisturbed soil samples collected in 100 cm³ steel cores (Kopecky rings) with 5.1 cm height and 5.0 cm diameter. The sampling was performed at four separate dates as shown in Supplement 5. Each sampling was done in six replicates, two per sub-plot (1 and 3, see Fig. 1), at two distinct depths in the upper subsoil (35 and 45 cm depth).

The cores were used to first determine water retention curves with a sandbox and pressure chambers according to the procedure outlined in Cornelis et al. (2005). The samples were brought to the matric suctions of 1, 3, 5, 7, 10, 33, 100 and 1500 kPa. Prior to moving the cores to the pressure chambers, air permeability (K_a) was determined at 10 kPa matric suction with the steady-state method of Grover (1955) according to the procedure described by Pulido Moncada et al. (2017). The used equipment consists of an annular water-filled reservoir with an open-bottomed float resting on the water. This creates an air chamber that goes through the inside of the water reservoir to the holding position of the sample. The air in the air chamber passes through the soil sample as the float falls. The falling rate of the float is a measure for the air permeability. This measurement was repeated for each sample at two different pressure gradients.

Air capacity (AC) and plant available water capacity (PAWC) were calculated from the water retention curve. AC represents the pore volume filled with air at field capacity, here considered at a matric suction of 10 kPa (pF 2.0) (White, 2006). PAWC, the ability of the soil to store and provide water to plant roots, is determined by subtracting the volumetric water content at field capacity by that at a matric suction of 1500 kPa (White, 2006). Bulk density (ρ_b) was determined by oven-drying the soil at 105 °C for 48 h (Grossman and Reinsch, 2002).

2.5. Crop parameters

Crop yields were determined annually in the three rotations from the start of the experiment. The exact harvest dates are presented in Supplement 6.

The maize crop yield was determined by manually harvesting two

adjacent rows (4 m length) of whole plants (cob and stem separately). This was repeated three times per sub-plot (1, 2 and 3, see Fig. 1), resulting in nine repeats per rotation-subsoiling combination.

The winter barley crop yield was determined with the help of a plot combine harvester with a width of 1.5 m. Per sub-plot three (1, 2 and 3, see Fig. 1) strips of 15 m length were harvested this way.

The alfalfa cuts (3–5 cuts per year) were harvested either with a small-scale harvester (Haldrup GmbH, Germany) or with a handheld cutter bar. These have a width of 1.5 m and 1.4 m, respectively. The choice of harvesting method was done based on the height of the alfalfa crop. The cutter bar was used when the crop grew taller. For each experimental sub-plot of alfalfa three (1, 2 and 3, see Fig. 1) strips of around 6 m (specified for each location) long was harvested per cut.

For each crop, sub-samples of 1 kg were collected to determine the oven-dry matter content (48 h at 70 °C).

Root number density and distribution were determined with the trench profile method (Van Noordwijk et al., 2001). Supplement 6 shows the frame (75 cm × 75 cm) that was used for this purpose. The frame was subdivided into smaller squares (2.5 cm × 2.5 cm) with strings. The use of this frame was previously described by Ren et al. (2019). First, an 80 cm deep trench profile was dug and the profile wall was carefully prepared with a knife to remove the smeared soil and make the roots more visible. After this preparatory stage, the number of roots was counted per smaller square. This observation was repeated on three dates (see Supplement 8). The proportion of roots in the subsoil was determined by dividing the number in the subsoil (30–75 cm) over the total number of roots in the frame.

2.6. Statistical analysis

All statistical analyses were performed with the software program SPSS Statistics 28 (SPSS, Chicago, USA). The physical soil parameters determined from the sampled Kopecky rings and the root parameters were grouped by subsoiling status, rotation and timing after subsoiling and/or sowing of a deep rooting (cover) crop (subsoiling × rotation × relative timing). Each grouping was regarded as a separate treatment in the further ANOVA analysis with block as a random factor. For the physical soil parameters the statistical analysis was performed separately for both sampling depths (i.e. 35 and 45 cm). If a significant effect was observed a Tukey post-hoc test was performed to determine significant differences ($P < 0.05$) between the different treatments/groupings. The statistical analysis of the root parameters was performed separately per crop (i.e. maize, fodder radish and alfalfa) to avoid comparing inherently different objects.

To reduce the effect of the strongly varying soil moisture content on soil strength (Bengough et al., 2011; Colombi et al., 2018) between the different measurement dates, the penetration resistance values (30–50 cm depth interval) obtained for the control sub-plots were consistently subtracted from those of the adjacent subsoiled sub-plots. In this way, we are looking to the change in penetration resistance relative to the control at different instants, which allows for comparison of values at different soil moisture conditions. Further analysis was performed between the three experimental rotations at similar timings relative to subsoiling. A similar procedure for processing and analysis was followed to evaluate the effects of fodder radish (rotation 2) and alfalfa (rotation 3) on the penetration resistance. Here the penetration resistance (30–50 cm depth interval) obtained for the control sub-plots of rotation 2 and rotation 3 were consistently subtracted from those of the control sub-plot of rotation 1 in the same block.

For crop yield the statistical analysis was limited to subsoiling as fixed factor and block as random factor per year. Only in 2021 were the results for maize yield compared between rotations 1 and 2.

3. Results

3.1. Soil physical parameters

Fig. 5-a presents the situation before the start of the experiment: a steep increase in penetration resistance starting at 30 cm depth (i.e. depth of ploughing), an extremely high peak (~6–7 MPa) around 35–40 cm depth, and a decrease to values that are to be expected for a pre-compacted sandy loam soil below 50 cm depth (Horn et al., 2007). Fig. 5-b, -c & -d display the average situation immediately after subsoiling for rotation 1, 2 and 3, respectively. The subsoiling of rotation 1 resulted in the disruption of the entire soil profile down to the cultivation depth of 50 cm (see Fig. 5-b: relatively homogeneous blue colour (i.e. < 4 MPa)). This was not the case for the subsoiling of rotations 2 and 3, where the number of tines was reduced from the original seven to four. In this instance the soil disturbance down to 50 cm depth was much more restricted to the pathways of the tines. The spacing between the tines (90 cm) becomes now observable on the contour plots. The regions between the pathways of the tines retained a relatively higher penetration resistance (See Fig. 5-c & -d: pathways of the tines are observable by the blue colour (i.e. < 4 MPa), while in between the pathways the colour gradually changes towards green and yellow (> 4 MPa)).

This initial difference in upper-subsoil (30–50 cm) disturbance between rotations 1, and rotation 2 and 3 is also evident when looking at the direct differences in penetration resistance between the adjacent subsoiled and control sub-plots. Fig. 6 presents these differences over time for all three experimental rotations, starting at the day of subsoiling. While the subsoiling of rotation 1 resulted in an average initial difference of -5.05 MPa, this was significantly lower for rotations 2 (-3.57 MPa – $P < 0.01$) and 3 (-3.05 MPa – $P < 0.01$). Starting from this initial difference the evolution of the difference in penetration resistance followed markedly different pathways. Rotation 1 saw a quicker reduction for the differences in penetration resistance than both rotation 2 and 3. After a similar number of primary field operations (see Supplement 1 – rotation 1: 2 × ploughing, 2 × harrowing, 2 × manure application and 1 maize harvest; rotation 2: 3 × ploughing, 3 × harrowing, 1 × manure application and 1 × maize harvest; rotation 3: 1 × ploughing, 1 × harrowing and 3 cuts of alfalfa), the status of both groups (rotation 1 – subsoiling under later, wetter soil conditions vs. rotations 2 and 3 – subsoiling under earlier, dry soil conditions) was completely inverted, with rotation 1 (average difference in penetration resistance 556 days after subsoiling: -0.73 MPa) having a significantly lower difference between the subsoiled and control sub-plots than rotation 2 (average difference in penetration resistance 438 days after subsoiling: -1.69 MPa – $P < 0.01$) and 3 (average difference in penetration resistance 438 days after subsoiling: -1.47 MPa – $P = 0.02$). The difference between rotation 2 and 3 at this point was insignificant ($P = 0.74$). The fitted logarithmic curves had a significant goodness of fit ($P < 0.01$) for all three rotations ($R^2 = 0.90$, $R^2 = 0.59$ and $R^2 = 0.59$, respectively).

The effects of the two deep-rooting crops on the penetration resistance of the upper-subsoil (30–50 cm depth) during the first two years after the sowing date is presented in Fig. 7. At none of the measurement dates did we observe a significant difference between the control sub-plots of rotations 2 and 3 and the control sub-plots of rotation 1. The difference in penetration resistance remained steadily around 0 MPa during the entire experimental period.

The other physical soil parameters, which were derived from laboratory analyses on the soil samples, are presented in Table 2 for both 35 cm and 45 cm depth. At 35 cm depth, the bulk density showed a similar trend for rotations 1 and 2. Starting from a very high value, the bulk density was reduced to less restrictive levels for root growth in a sandy loam soil (< 1.63 g cm⁻³; Goldsmith et al., 2001; USDA, 1999), although, only for rotation 2 the reduction was significant ($P = 0.01$). On the longer-term, the reduction was again completely reversed for both rotations. For rotation 3, no observation for the short-term is available. However, it is clear that the longer-term result is an

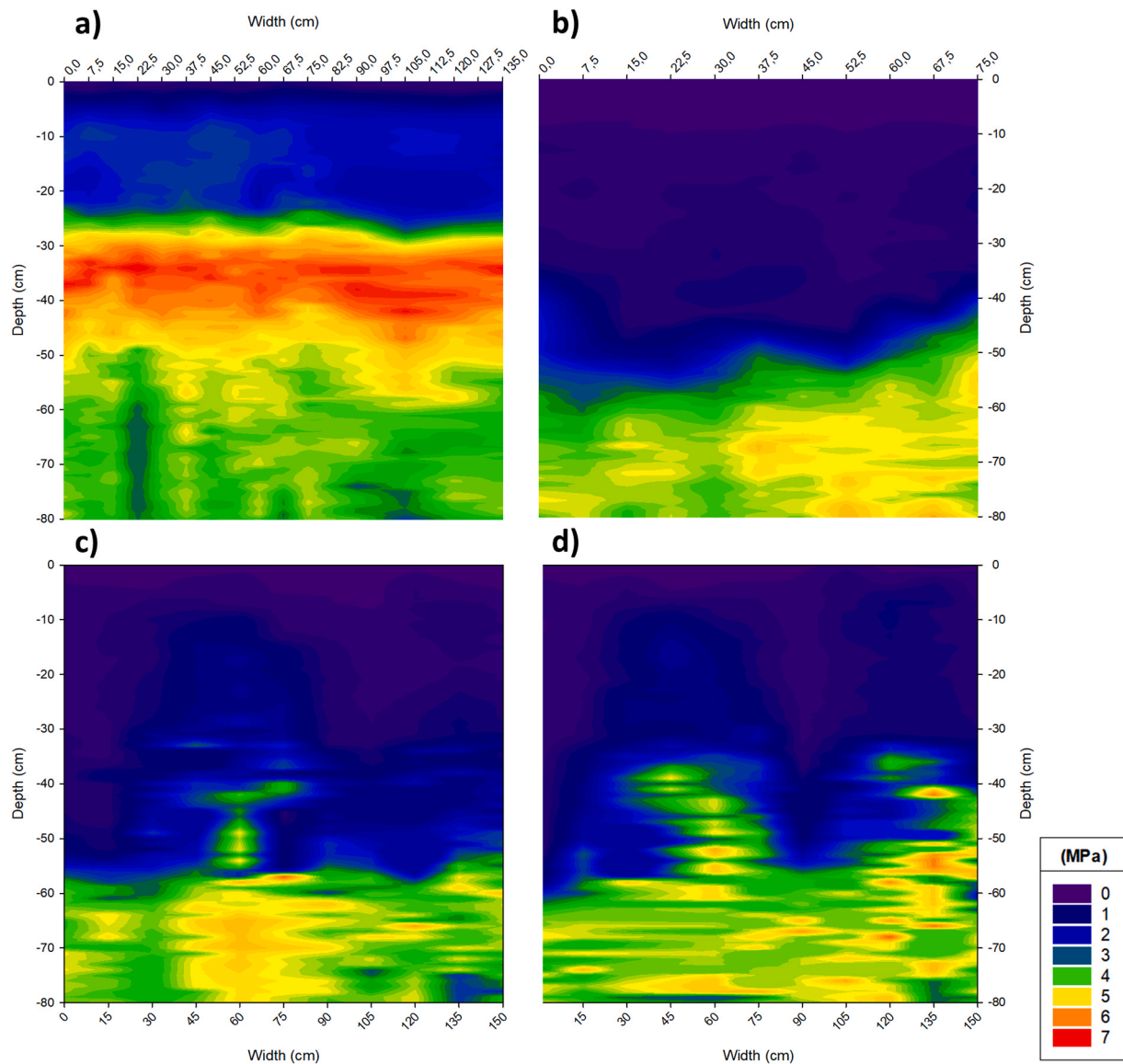


Fig. 5. Average contour plots of penetration resistance perpendicular to the cultivation direction ($n = 6 - \text{MPa}$) of the control sub-plots of rotation 1 (16/04/2020) (a), the subsoiled sub-plots of rotation 1 right after subsoiling (6/11/2019) (b), the subsoiled sub-plots of rotation 2 right after subsoiling (3/09/2020) (c) and the subsoiled sub-plots of 3 right after subsoiling (3/09/2020) (d). The markers on the x-axis pinpoint the relative locations of the distinct measurements.

insignificant ($P = 0.99$) increase compared to the reference condition in the control sections of rotation 1. None of the control sections of rotations 2 ($P = 0.99$) and 3 ($P = 0.99$) resulted in a significant reduction after sowing a deep rooting (cover) crop neither.

A similar trend can be observed for porosity. Unexpectedly, none of the experimental treatments had a significant effect on the AC of the upper-subsoil. All treatments displayed values well below the threshold value for ample aeration (i.e. $0.10 \text{ m}^3 \text{ m}^{-3}$; White, 2006). There was also no significant impact on the air permeability. This parameter did show very high variability within the different sampling groupings. Only two groupings came close to the limiting value for good aeration of $20 \mu\text{m}^2$, the subsoiled sub-plots of rotation 1 after 230 days and of rotation 2 after 461 days of subsoiling. Many of the measured samples were almost completely impermeable for air ($<1 \mu\text{m}^2$) (Ball et al., 1988; Fish and Koppi, 1994). PAWC showed significant ($P = 0.02$) differences between the sampling groupings. However, this was limited to a significant difference between the subsoiled section of rotation 1 after 604 days of subsoiling and the subsoiled sections of rotation 2 after 190 days of subsoiling and marginally lower ($P = 0.07$) PAWC with the control

section of rotation 2 after 456 days of sowing fodder radish. Almost all measured samples had an optimal water holding capacity ($>0.20 \text{ m}^3 \text{ m}^{-3}$; Cockroft and Olsson, 1997; Hall et al., 1977; Verdonck et al., 1984; White, 2006).

At 45 cm depth the effects of subsoiling and rotation seemed much more limited. No significant reductions were observed for bulk density ($P = 0.59$), porosity ($P = 0.71$), and PAWC ($P = 0.35$). Only AC ($P < 0.01$) and, to a lesser extent ($P = 0.06$), air permeability seemed impacted by the experimental treatments. The highest values for AC were found relatively shortly after subsoiling, while the lowest values were observed for rotation 3 and the control sub-plots of rotation 2. Just as at 35 cm depth, there were no statistically significant differences in air permeability, also resulting from the high levels of variability. The relatively high average value for the subsoiled sub-plots of rotation 2 after 461 days of subsoiling was caused solely by one sample with a well-developed biopore ($59.82 \mu\text{m}^2$). The other samples of this group were close to or below the limit for impermeability ($<1 \mu\text{m}^2$) (Ball et al., 1988; Fish and Koppi, 1994). The samples of rotation 3 also showed clear biopores (See example in Supplement 9). However, these were still

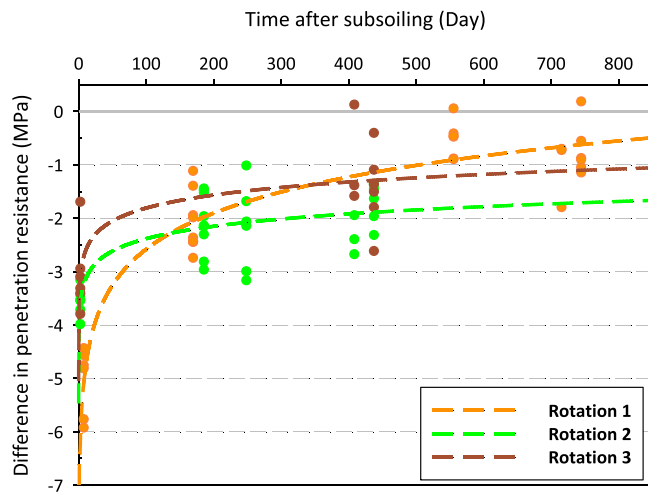


Fig. 6. Difference in average penetration resistance (MPa) at 30–50 cm depth between adjacent subsoiled and control sub-plots over time after subsoiling for rotation 1 (orange), rotation 2 (green) and rotation 3 (brown). The circles indicate the repeats per measurement date (2 per block at locations 1 and 3; see Fig. 1). The dashed lines indicate the logarithmic curve fitted in SPSS Statistics 28 (rotation 1: $R^2 = 0.90$; rotation 2: $R^2 = 0.59$; rotation 3: $R^2 = 0.59$).

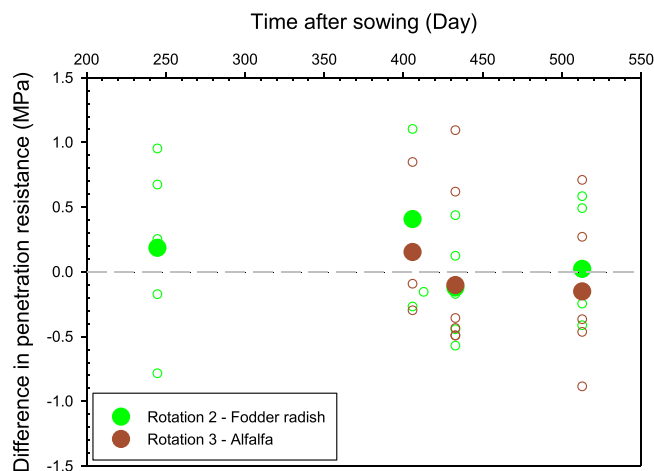


Fig. 7. Difference in average penetration resistance (MPa) at 30–50 cm depth between the control sub-plots of both rotation 2 (green) and rotation 3 (brown) and the control sub-plots of rotation 1 in the same block and location (1 and 3, respectively; see Fig. 1). The smaller, unfilled circles indicate the repeats ($n = 6$; expect at 406 days after sowing, when $n = 3$). The bigger, filled circles indicate the overall averages.

filled with alfalfa roots at the time of measurement.

3.2. Crop parameters

3.2.1. Crop yield

In general, subsoiling resulted in higher crop yields in all three crop rotations (Table 3), but no significant differences were observed. The crop response to subsoiling did show a clear difference between the years of the experimental period. The maize crop yield of rotation 1 was positively impacted by subsoiling in 2020 (+8% - $P = 0.08$) and in 2022 (+8% - $P = 0.19$), but was slightly reduced in 2021 (-3% - $P = 0.61$). The most notable difference that was observed in this intermediate year was not solely caused by subsoiling, but by a combination of rotation and timing after subsoiling (rotation 1: 701 days after subsoiling vs. rotation 2: 390 days after subsoiling). The maize crop yield in rotation 2 was substantially higher (+17% - $P = 0.06$) than in rotation 1.

The effects of subsoiling are further visualized in Fig. 8, which presents the average and per block yield gains for all rotations and makes clear the large variation in the crop response between the experimental blocks.

3.2.2. Root growth

Table 4 presents the topsoil and subsoil root number density and the proportion of roots in the subsoil. The statistical analysis was performed separately for the observed (cover) crops (i.e. maize, fodder radish and alfalfa). Statistically significant differences were only observed for maize. The topsoil root number density of rotation 1 was significantly ($P < 0.01$) higher in 2020 than in 2021. Other, less obvious, observations for the topsoil root number density were the initial reduction (-21% - $P = 0.49$) after subsoiling in rotation 1, which was no longer the case in 2021, and the insignificant increase (+51%) in rotation 2 compared to rotation 1 in 2021. The differences for the subsoil root number density were more substantial, however, the relative increase in standard deviation led to only marginal statistical differences. The subsoil root number density of maize in rotation 1 increased by 523% after subsoiling ($P = 0.09$), but after one year the advantage was already reduced to +103% ($P = 0.99$). For maize in rotation 2, this effect was even smaller (+51% - $P = 0.97$). The control sub-plots of rotation 2 had a 229% higher subsoil root number density than rotation 1 in 2021, but this difference remained insignificant ($P = 0.90$). The proportion of roots in the subsoil for the control sub-plots of the maize crop of rotation 1 was extremely low in 2020 (1.6%) and was only slightly better in 2021 (6.2% - $P = 0.67$). For rotation 2, control plots had a slightly higher proportion in the subsoil (9.6%). This rotation effect is visualized in Fig. 9-b. Subsoiling significantly ($P = 0.05$) increased the proportion of roots in the subsoil in 2020–12.2%, but this beneficial effect did not last until 2021 (10.8% - $P = 0.66$). For rotation 2 the beneficial effect did not even materialize in the first year (i.e. 2021 - $P = 0.75$). The effect of subsoiling on the subsoil root growth (+231.0%) is visualized in Fig. 9-a and it highlights the large variation in subsoil root growth between the blocks. The proportion of subsoil root growth for all maize observations gradually, but significantly ($P < 0.01$), declined from block A ($14.98 \pm 7.7\%$) over B ($7.2 \pm 3.9\%$) to C ($4.9 \pm 3.5\%$). In the most northerly located sub-plots of this block (i.e. subsoiled rotation 2) we observed an extremely high groundwater level (around 70 cm depth) when digging the profile trenches (Supplement 10-a). 11 m further to the south in the control sub-plot, no such observation was made (Supplement 10-b). Yet, mottling pointed towards similar anaerobic conditions in the past. This gradient was even more substantial ($P < 0.01$) when we only took the subsoiled sub-plots into account: Block A ($20.6 \pm 2.7\%$) over B ($9.2 \pm 0.5\%$) to C ($6.9 \pm 2.7\%$). The gradient was similar for fodder radish and alfalfa.

Overall, fodder radish had the highest topsoil and subsoil root number density, but the proportion of roots in the subsoil was of a similar level to alfalfa. The subsoil root growth of fodder radish (+61.9%) and alfalfa (+36.4%) were least impacted by subsoiling, primarily because the proportion of roots in the subsoil was so high for the control sub-plots. Both crops showed clear signs of preferential growth down into the pathways of the tines and not uniformly throughout the soil profile (Supplement 11).

4. Discussion

By incorporating subsoiling operations and utilizing bio-subsoilers like fodder radish and alfalfa within feasible fodder crop rotations, we ensured practical applicability of the experimental results for farmers. However, altering the crop rotation necessitated adjustments in farming practices and/or their timing. Consequently, inconsistencies between the tested crop rotations were inevitable, both in the timing of subsoiling (rotation 1: 29 October 2019; rotations 2 and 3: 31 August 2020 under significantly drier conditions) and in the disruption intensity, because of the removal of the three front tines of the subsoiler (standard: seven

Table 2
Average bulk density (ρ_b), porosity (Φ), air capacity (AC), plant available water capacity (PAWC) and air permeability (K_a) $\pm$ standard deviation ($n = 6$) at 35 cm and 45 cm depth for the control and subsoiled sub-plots of rotations 1, 2 and 3 at a specified number of days after subsoiling (DaS) and sowing of the deep-rooting crop or bio-subsoiler (DaB). Different letters indicate significant differences ($P < 0.05$) for the Tukey post-hoc test, n.s. points out insignificant ($P > 0.10$) differences.

Rotation	Subsoiling	DaS (days)	DaB (days)	ρ_b (g cm ⁻³)		Φ (-)		AC (m ³ m ⁻³)		PAWC (m ³ m ⁻³)		K_a (μ m ²)	
				35 cm	45 cm	35 cm	45 cm	35 cm	45 cm	35 cm	45 cm	35 cm	45 cm
Rotation 1	Control	-	-	1.67 $\pm$ 0.04 ab	1.69 $\pm$ 0.05 n.s.	0.37 $\pm$ 0.02 abc	0.36 $\pm$ 0.02 n.s.	0.06 $\pm$ 0.02 n.s.	0.06 $\pm$ 0.01 abc	0.23 $\pm$ 0.02 ab	0.21 $\pm$ 0.02 n.s.	1.36 $\pm$ 0.74 n.s.	6.73 $\pm$ 8.86 ab
Rotation 1	Subsoiled	230	-	1.60 $\pm$ 0.04 ab	1.69 $\pm$ 0.08 n.s.	0.40 $\pm$ 0.01 bc	0.36 $\pm$ 0.03 n.s.	0.07 $\pm$ 0.02 n.s.	0.07 $\pm$ 0.01 c	0.23 $\pm$ 0.02 ab	0.20 $\pm$ 0.02 n.s.	14.68 $\pm$ 20.95 n.s.	2.78 $\pm$ 1.20 ab
Rotation 1	Subsoiled	604	-	1.66 $\pm$ 0.03 ab	1.68 $\pm$ 0.03 n.s.	0.37 $\pm$ 0.01 abc	0.37 $\pm$ 0.01 n.s.	0.07 $\pm$ 0.01 n.s.	0.06 $\pm$ 0.01 abc	0.20 $\pm$ 0.01 a	0.20 $\pm$ 0.02 n.s.	2.54 $\pm$ 2.73 n.s.	8.45 $\pm$ 8.56 ab
Rotation 2	Control	-	184	1.70 $\pm$ 0.05 b	1.70 $\pm$ 0.03 n.s.	0.36 $\pm$ 0.02 ab	0.36 $\pm$ 0.01 n.s.	0.05 $\pm$ 0.02 n.s.	0.05 $\pm$ 0.01 ab	0.21 $\pm$ 0.01 n.s.	0.20 $\pm$ 0.01 n.s.	4.29 $\pm$ 4.96 n.s.	3.14 $\pm$ 2.50 ab
Rotation 2	Control	-	456	1.65 $\pm$ 0.04 ab	1.73 $\pm$ 0.04 n.s.	0.38 $\pm$ 0.02 abc	0.35 $\pm$ 0.02 n.s.	0.04 $\pm$ 0.02 n.s.	0.04 $\pm$ 0.01 ab	0.24 $\pm$ 0.02 ab	0.22 $\pm$ 0.01 n.s.	6.98 $\pm$ 9.05 n.s.	2.87 $\pm$ 4.68 ab
Rotation 2	Subsoiled	190	184	1.57 $\pm$ 0.03 a	1.68 $\pm$ 0.06 n.s.	0.41 $\pm$ 0.01 c	0.37 $\pm$ 0.02 n.s.	0.06 $\pm$ 0.01 n.s.	0.06 $\pm$ 0.01 bc	0.25 $\pm$ 0.02 b	0.23 $\pm$ 0.05 n.s.	5.20 $\pm$ 5.74 n.s.	3.43 $\pm$ 2.26 ab
Rotation 2	Subsoiled	461	456	1.68 $\pm$ 0.07 b	1.68 $\pm$ 0.02 n.s.	0.36 $\pm$ 0.03 ab	0.37 $\pm$ 0.01 n.s.	0.06 $\pm$ 0.01 n.s.	0.05 $\pm$ 0.01 abc	0.23 $\pm$ 0.03 ab	0.22 $\pm$ 0.02 n.s.	14.56 $\pm$ 25.90 n.s.	15.50 $\pm$ 23.64 b
Rotation 3	Control	-	456	1.65 $\pm$ 0.05 ab	1.71 $\pm$ 0.05 n.s.	0.38 $\pm$ 0.02 abc	0.36 $\pm$ 0.02 n.s.	0.05 $\pm$ 0.01 n.s.	0.04 $\pm$ 0.01 ab	0.23 $\pm$ 0.02 ab	0.22 $\pm$ 0.01 n.s.	4.92 $\pm$ 4.03 n.s.	0.93 $\pm$ 0.97 ab
Rotation 3	Subsoiled	461	456	1.70 $\pm$ 0.07 b	1.70 $\pm$ 0.05 n.s.	0.36 $\pm$ 0.03 a	0.36 $\pm$ 0.02 n.s.	0.05 $\pm$ 0.01 n.s.	0.04 $\pm$ 0.02 a	0.22 $\pm$ 0.02 ab	0.21 $\pm$ 0.01 n.s.	2.21 $\pm$ 1.41 n.s.	0.67 $\pm$ 0.75 a
Statistical analysis				P-value									
subsoiling x rotation x relative timing				< 0.01	0.59	< 0.01	0.71	0.11	< 0.01	0.02	0.35	0.73	0.06

Table 3
Average dry matter crop yield (kg ha⁻¹) $\pm$ standard deviation ($n = 9$) for control and subsoiled sections of the tested rotations during the experimental period (2020–2022).^a.

Crop yield (kg ha ⁻¹)		2020	2021	2022
Rotation 1	Crop	Maize	Maize	Maize
	Control	15992 $\pm$ 885	21087 $\pm$ 2465	15544 $\pm$ 2065
	Subsoiled	17216 $\pm$ 2092	20475 $\pm$ 2225	16722 $\pm$ 1506
Rotation 2	Crop	Winter wheat	Maize	Winter barley
	Control	-	24223 $\pm$ 2264	6646 $\pm$ 422
	Subsoiled	-	24434 $\pm$ 2227	6909 $\pm$ 386
Rotation 3	Crop	Winter wheat	Alfalfa	Alfalfa
	Control	-	13410 $\pm$ 692	17026 $\pm$ 1370
	Subsoiled	-	13898 $\pm$ 842	17392 $\pm$ 744

^a No significant differences ($P < 0.05$) were observed during the statistical analysis.

tines) for rotations 2 and 3. These inconsistencies were considered in the evaluation of the experimental results.

4.1. Mechanical remediation

Prior to the establishment of the experiment a strongly compacted layer had formed at the interface between the top- and subsoil (30–50 cm). The extremely high mechanical impedance of this layer resulted in an almost complete inhibition of subsoil root exploration. In the relatively dry year of 2020, only 1.6% of the maize root number was observed below 30 cm depth. The level of root restriction did seem to depend on the soil moisture content, because in the relatively wet year of 2021 it was increased, although insignificantly, up to 6.2%. Schittenhelm and Schroetter (2014) observed a similar proportion (7%) of maize root length density in the subsoil under non-compacted wet soil conditions. Under non-compacted dry soil conditions the proportion of

the maize root length density in the subsoil increased to 18%. The difference in crop response between our results and the results by Schittenhelm and Schroetter (2014) could exemplify the aggravated impact of soil compaction under extreme soil moisture conditions (Benjamin et al., 2003; Lindstrom and Voorhees, 1994). While a non-restricted maize plant would be stimulated to explore the subsoil under dry soil conditions, the same moisture conditions will make an already compacted soil layer even more difficult to penetrate (Colombi et al., 2018). However, the insignificance of these observations and the absence of this phenomenon for the absolute root numbers make this reasoning less conclusive. The dependence of crop response to the level of compaction on weather conditions was also reflected in crop yield. Subsoiling resulted in an 8% increase in maize dry matter yield in the dry years of 2020 and 2022. In the much wetter year of 2021, the subsoiled sub-plots even had a 3% reduction in maize yield. The importance of dry spells to increase the potential benefits of subsoiling has been widely observed in previous field studies (Olesen and Munkholm, 2007; Schneider et al., 2017).

Subsoiling has a very high draft requirement (Kichler et al., 2011) and this is magnified when the operation is performed at dry soil conditions (Raper and Sharma, 2004). The conditions at the end of August 2020 proved too dry for the used tractor to pull the subsoiler with seven tines down to a depth of 50 cm. After removal of the three frontal tines, subsoiling could be successfully done for rotations 2 and 3. The result was however a significantly reduced soil disruption compared to the subsoiling in rotation 1, a similar result as had been observed for different tine spacings by Vanderhasselt et al. (2023). The higher lateral variability in soil disruption remained observable for a long time after subsoiling in both penetration resistance and root growth. The roots grew preferentially through the pathways of the tines (e.g. Supplement 11), which is comparable to the preferential growth of roots into artificial macropores in otherwise compacted subsoils, as observed by Stirzaker et al. (1996), Colombi et al. (2017a) and Atkinson et al. (2019).

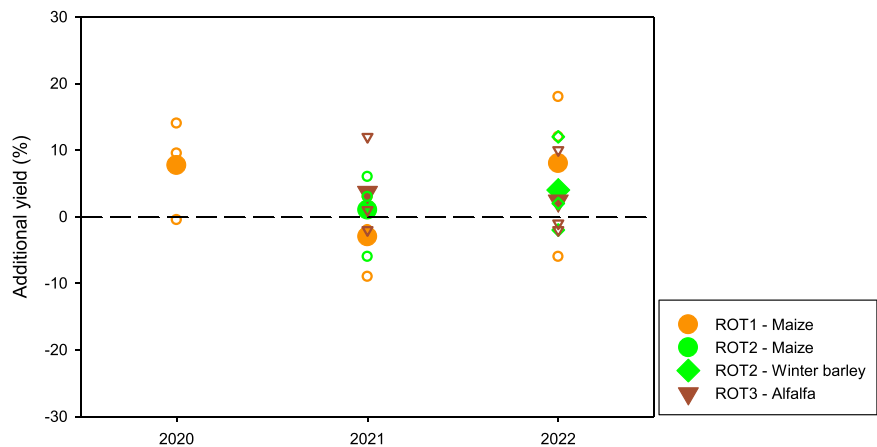


Fig. 8. Additional yield (subsoiling-control/control x 100; %) for rotations 1 (orange), 2 (green) and 3 (brown). The larger, coloured-in symbols give the average value per rotation. The smaller, blank symbols give the value per block. The symbol itself describes the crop of interest: maize (circle), winter barley (diamond) and alfalfa (triangle).

Table 4

Average topsoil and subsoil root number density (# roots m⁻²) and proportion of subsoil root number compared to total (0–75 cm) root number (%) ± standard deviation (n = 3) for the experimental rotations at different timing (days) after subsoiling (DaS) and the sowing of deep rooting crops or bio-subsoilers (DaB). Different letters indicate significant differences (P < 0.05) for the Tukey post-hoc test, n.s. points out insignificant (P > 0.10) differences.

Rotation	Subsoiling	Crop	Observation date	DaS (days)	DaB (days)	Topsoil (# roots m ⁻²)	Subsoil (# roots m ⁻²)	Proportion roots in subsoil (%)
Rotation 1	Control	Maize	15/09/2020	-	-	3183.7 ± 659.9	39.5 ± 32.2	1.6 ± 1.1
						b	a	a
Rotation 1	Control	Maize	14/10/2021	-	-	736.3 ± 121.8	30.6 ± 7.0	6.2 ± 2.1
						a	ab	ab
Rotation 1	Subsoiled	Maize	15/09/2020	322	-	2518.5 ± 175.3	245.9 ± 160.8	12.2 ± 7.4
						b	a	b
Rotation 1	Subsoiled	Maize	14/10/2021	716	-	743.7 ± 106.3	62.2 ± 35.6	10.8 ± 4.9
						a	a	ab
Rotation 2	Control	Maize	14/10/2021	-	405	1158.5 ± 368.9	100.7 ± 89.6	9.6 ± 6.2
						a	a	ab
Rotation 2	Subsoiled	Maize	14/10/2021	409	405	1080.0 ± 613.2	152.1 ± 150.3	13.7 ± 5.9
						a	a	b
Statistical analysis						P-values		
subsoiling x rotation x relative timing						< 0.01	0.07	0.02
Rotation 2	Control	Fodder radish	02/12/2020	-	89	4768.9 ± 1359.1	558.0 ± 191.6	14.7 ± 1.4
						n.s.	n.s.	n.s.
Rotation 2	Subsoiled	Fodder radish	02/12/2020	93	89	4035.6 ± 790.7	936.3 ± 424.3	24.1 ± 6.5
						n.s.	n.s.	n.s.
Statistical analysis						P-values		
subsoiling x rotation x relative timing						0.29	0.16	0.13
Rotation 3	Control	Alfalfa	14/10/2021	-	405	1163.0 ± 246.6	130.4 ± 52.3	14.4 ± 5.3
						n.s.	n.s.	n.s.
Rotation 3	Subsoiled	Alfalfa	14/10/2021	409	405	1275.6 ± 351.3	190.6 ± 108.4	17.1 ± 4.1
						n.s.	n.s.	n.s.
Statistical analysis						P-values		
subsoiling x rotation x relative timing						0.42	0.58	0.68

Unlike for penetration resistance, the subsoiling did not result in significant improvements of the parameters (i.e. bulk density, porosity, AC, PAWC and air permeability) obtained from the soil samples down to the working depth (i.e. 50 cm). Only the upper-section of the compacted soil layer (35 cm depth) was significantly loosened at the time of the first sampling. It seems that the lowest reaches of the subsoiling operation (45 cm) were recompacted extremely fast or there was no loosening until this depth, only temporary disruption (Fig. 5). The high values for PAWC in the extremely compacted subsoil layer highlight the relative importance of accessibility, referring to the ability of plant roots to reach plant available water (Bengough et al., 2011) over availability, which pertains to the proportion of soil water that can potentially be extracted by plant roots (Bengough et al., 2011)), as was previously described by Bouma (1990) and Colombi et al. (2018). The upper-subsoil of our experimental field had an optimal level of plant available water

(Cockroft and Olsson, 1997; Hall et al., 1977; Verdonck et al., 1984; White, 2006). However, it was probably only of use when it became accessible through deep tillage, which might explain the increase in maize yield after subsoiling in dry years in rotation 1.

The increased accessibility after subsoiling is evidenced by the marked increase in subsoil root growth after subsoiling. This was especially the case for the maize crop of rotation 1 in the dry year of 2020. Maize, being relatively sensitive to soil compaction (Lipiec and Simota, 1994), had a bigger crop response towards subsoiling compared to tap-rooted fodder radish and alfalfa, which have a high capacity for penetrating compacted soil layers (Chen and Weil, 2010; Han et al., 2015; Kautz et al., 2010; Pulido-Moncada et al., 2020; Uteau et al., 2013). A similar increase in subsoil root exploration of maize after subsoiling was observed by Varsa et al. (1997) on a silt loam soil. The presence of a restricting layer not only resulted in a reduced root

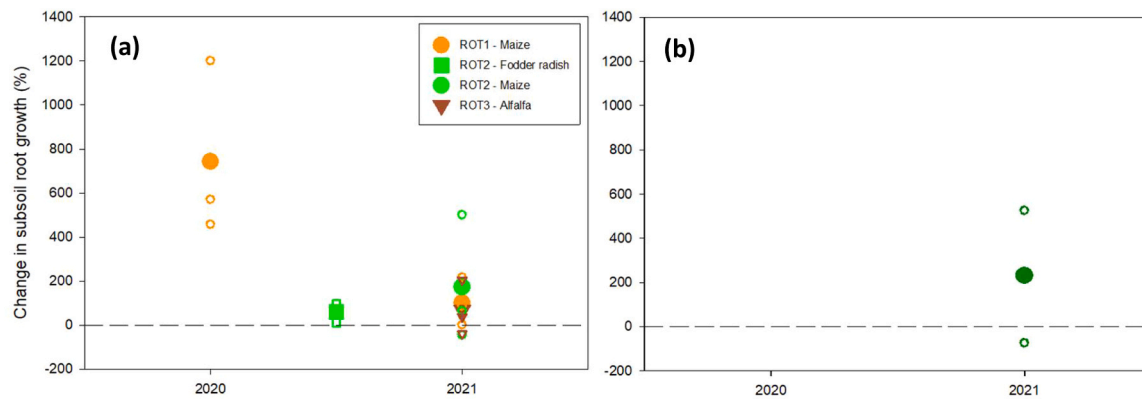


Fig. 9. Proportional change in subsoil root growth (%) from control to subsoiled sub-plots (a) and from maize in the control sub-plots of rotation 1 to maize in the control sub-plots of rotation 2 (b). Rotation 1 is indicated in orange, rotation 2 in green and rotation 3 in brown, while the difference between rotation 1 and 2 is indicated in dark green. The larger, coloured-in symbols give the average value per rotation. The smaller, blank symbols give the value per block. The symbol itself describes the crop of interest: maize (circle), fodder radish (square) and alfalfa (triangle).

exploration of the subsoil, it also led to an increase in the topsoil root growth. Subsoiling reduced the topsoil root number density of both maize and fodder radish, while increasing the root number density in the subsoil. This sort of compensatory root growth (Correa et al., 2019; Lynch et al., 2021; Lynch and Wojciechowski, 2015) was previously observed by Ehlers et al. (1983) and Tardieu (1988) for oat and maize, respectively. The perennial crop, alfalfa, did not show any sign of this process, although we only measured root density in the first year after sowing, while it is expected that alfalfa only reaches its full potential after minimal 2 years in the field (Lamb et al., 2000).

The changes in root distribution by subsoiling were, however, highly variable within the experimental field, which could be linked to the clear gradient in overall subsoil root growth. The proportion of subsoil root growth decreased from block A to C. In block C the disruption of the restricting subsoil layer by subsoiling hardly led to an increase in deeper root exploration, especially for rotation 2 and 3, the most northerly plots of the block. This is most likely linked to the existence of a gradient in drainage capacity and groundwater level within the experimental field, which would explain the gradient in the ECa values measured with the EMI soil scan (Brevik et al., 2006). The subsoil conditions of block C seem to have been so limiting (see Table 1 and Supplement 10 – presence of Cg subsoil horizon, indicating periodic anaerobic conditions) that even without the restricting layer, deeper root exploration would not compensate for its high energetic investment (Correa et al., 2019; Lambers et al., 2002; Lynch, 2007; Lynch et al., 2021).

4.2. Biological remediation

Unlike subsoiling, fodder radish and alfalfa were not able to significantly decrease the penetration resistance (Fig. 7) or bulk density (Table 2) of the compacted subsoil layer. A similar result was obtained in the study of Guaman et al. (2016), which compared the remediating effect of several radish species with inter-row subsoiling on penetration resistance and potato crop growth on a loamy sand soil. Abdollahi and Munkholm (2014) on the other hand did observe a reduction in the penetration resistance of a plough pan at the interface between top- and subsoil, but only after five consecutive years of fodder radish as cover crop.

Both fodder radish and alfalfa clearly penetrated the highly compacted subsoil layer. This was evidenced by the presence of biopores inside the sampled Kopecky rings (e.g. Supplement 9) and the markedly higher subsoil root growth compared to maize in the control sub-plots. The relatively high levels of root penetration into the compacted subsoil corresponds with previous field experiments (Chen and Weil, 2010; Han et al., 2015; Kautz et al., 2010; Pulido-Moncada et al., 2021b; Uteau et al., 2013).

The created biopores did not significantly improve the pore functionality of the compacted subsoil layer, in contrast to the observations of Pulido-Moncada et al. (2021) for alfalfa and Chen et al. (2014) for fodder radish. For alfalfa the low levels of air permeability could at least partly be explained by the presence of the non-decayed roots inside biopores. Air permeability is highly responsive to macroporosity and pore continuity, which would increase with the presence of biopores (Cavaliere et al., 2009; Dörner and Horn, 2006). However, as Barley (1954) noted previously, air permeability will only increase as soon as the original roots are gone. The noteworthy increase in the variability of air permeability after fodder radish could be interpreted as an increase in structural elements, like biopores. A higher number of samples might have been required to account for the large in-plot variability. A more thorough investigation of the effects of the potential biopores by alfalfa and fodder radish on the rooting capacity of the subsequent crop, will require the additional assessment of a uniform crop for all experimental crop rotations after the period discussed here.

The maize yield in 2021 was notably higher for rotation 2 than for rotation 1. Breaking the maize monoculture with a winter cereal and fodder radish (cover) crop had a positive impact on the maize crop. The average dry matter yield increased by 17%, while the topsoil and subsoil root number density increased by 51% and 231%, respectively. The beneficial impact of breaking arable crop monocultures with crop rotations has been widely demonstrated in previous long-term studies. Nevens and Reheul (2002, 2001) observed higher crop yields for forage maize in rotation compared to monoculture and linked this primarily to a nitrogen contribution effect. In our experiment the additional nitrogen contribution originated from the mineralization of the grain stubble and fodder radish. This probably gave rotation 2 a relative advantage to rotation 1, since both rotations had received the same, relatively low, amount of fertilizer in 2021. Additional causes of the rotation effect have been found in improvements to soil moisture content, soil fungal and microbial populations, soil-borne disease pressure and soil structure (Blanco-Canqui et al., 2015; Van Zeeland et al., 1999; Weisskopf et al., 1995). The relatively high, although insignificant, increase in subsoil root growth does seem to imply that the previously inaccessible soil layer was made more accessible by breaking the maize monoculture. Recycling of the biopores created by fodder radish could help to explain this observation. The high number of fodder radish roots in the subsoil does point towards this possibility, although no systematic determination was done of the number of biopores. Chen and Weil (2011) and Guaman et al. (2016) observed a similar increase in maize subsoil root growth following radish species cover crops. Biopores are more stable than interaggregate pores and could be reused for several years (Hagedorn and Bundt, 2002; Schäffer et al., 2007). The eventual loss in biopore frequency will most likely be caused by tillage (Or et al., 2021). The

relatively wet growing season in 2021 makes it less likely that the stimulated deeper root exploration for the 2021 maize crop in rotation 2 had a substantial effect on the crop yield. As for subsoiling (Olesen and Munkholm, 2007; Schneider et al., 2017), increased accessibility to the subsoil will mainly have a beneficial impact in growing seasons with prolonged dry spells, which was not the case in 2021. The nitrogen contribution effect can probably explain most of the observed rotation effect as all crop rotations received the same amount of mineral N irrespective of the soil N supply.

4.3. Recompaction

The subsoiled sub-plots for all three experimental crop rotations were recompacted extremely fast in the first year after the soil disruption. This is highlighted by the very good fit of the logarithmic curves, especially for rotation 1. The high risk of recompaction following subsoiling has been widely remarked (Canarache et al., 2000; Munkholm et al., 2005; Spoor et al., 2003). On a slightly longer time frame differences became noticeable between the disruption status (here quantified by the difference in penetration resistance between the adjacent subsoiled and control sub-plots) of the experimental crop rotations. Even though rotation 1 started with a significantly higher disruption of the upper subsoil, it did not take a long time to recompact to similar levels as rotation 2. By the start of the second year after subsoiling, rotation 1 was even significantly more recompacted than both rotation 2 and 3, even though the frequency of field operations was not limited in the latter two rotations. Limiting traffic intensity has been demonstrated to reduce recompaction (Elsen et al., 2014; Munkholm et al., 2005; Soane et al., 1986). The difference between the rotations could be found in the reduced initial subsoil disruption and the presence of deep rooting (cover) crops. Although there was no significant difference in recompaction between both crops, fodder radish did show a marginally better stabilization of the upper-subsoil than the slower growing alfalfa (Lamb et al., 2000). This might be explained by the relatively fast and expanded root exploration of fodder radish. The cover crop had a markedly higher subsoil root number density after a limited growing period of 3 months than alfalfa had after 1 year without soil tillage. The combination of limiting soil disruption (here by subsoiling with only 4, instead of 7 tines) and sowing deep rooting (cover) crops seems to have been a successful measure to prolong the impact of subsoiling and keep the subsoil accessible for deeper root exploration. The importance of initial soil disruption during subsoiling on the rate of recompaction has been previously remarked by Vanderhasselt et al. (2023). In this study the 7-tine subsoiler disrupted the subsoil to a higher degree than the 3- and 5-tine subsoilers, but also led to a significantly quicker recompaction.

Despite the clear effects on penetration resistance, no similar trends were observed for bulk density and porosity. Both parameters showed a clear improvement shortly after subsoiling for rotations 1 and 2 and a complete recompaction after one to two years' time for all three.

5. Conclusion

This experiment demonstrated that subsoiling is an effective remediation measure to disrupt heavily compacted layers in the upper-subsoil, at least in the short-term. The crop response was however highly variable and was only positive when the opportunity of reaching the subsoil proved to be advantageous for the crop (i.e. dry season). Besides its high degree of variability between years, the effects also proved to be very short-lived under a standard arable fodder crop rotation. Including deep rooting (cover) crops, like fodder radish and alfalfa, and limiting the initial subsoil disruption helped to partially stabilize the disrupted subsoil and reduce recompaction. The difference between both crops was limited. Despite alfalfa's longer growing period and the limit of tillage during its cultivation, the quicker growing fodder radish had a slightly better effect in this respect.

Fodder radish and alfalfa penetrated the compacted subsoil without

previous disruption by subsoiling. However, this did not lead to a significant improvement of the measured soil physical parameters.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.still.2023.105956](https://doi.org/10.1016/j.still.2023.105956).

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